UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2023/2024 SECOND SEMESTER EXAMINATION

COURSE: MTH324: COMPLEX VARIABLES II
DATE SCHEDULED: THURSDAY 17th JULY, 2025
TIME ALLOWED: TWO HOURS (2 HRS)

Instructions: Attempt any four (4) questions only.

1. (a) State the Laurent's theorem, given that $f(z)$ is a complex function
(b) (i) State the maximum modulus theorem (MMT).
If $f(z)$ is analytic every where in the complex plane, using the MMT, show (use the function, $f(z) = \sin z$) that the maximum modulus theorem depends on the domain of the function and the topology of the region.
(ii) State the Rouche's theorem.

2. (a) State the necessary and sufficient conditions for a function, $f(z)$ to be analytic.
(b) Find the Poles and Zeros of $f(z) = (\frac{z-1}{z^2})\sin(\frac{1}{z-2})$

3. (a) Define (i) Analytic function, (ii) Singular point of a function, $f(z)$.
(b) Locate, in the finite z-plane, all the singularities, if any, and name them:
(i) $f(z) = \dfrac{2z^2 - z - 1}{(z-3)^3(z-i)(z+1-2i)}$
(ii) $f(z) = \dfrac{\sin mz}{z^2 + 2z + 2}$, $mz > (0)$.
(iii) $f(z) = \dfrac{1}{z^4 + 1}$

4. (a) (i) Define removable singularity of a single-valued function $f(z)$.
(ii) Show that $f(z) = \dfrac{\sin(z-a)}{z-a}$ has a removable singularity at $z = a$
(b) Evaluate $\dfrac{1}{2\pi i}\int_c \dfrac{\exp(zt)}{z^2(z^2+2z+2)}dz$ using the residue theorem, where $c : |z| = 3$

5. (a) State the Cauchy's residue theorem.
(b) Evaluate the integral, using the residue theorem,
$\int_c \dfrac{5z-2}{z(z-1)}dz$, where $c : |z| = 2$.

6. (a) If $f(z)$ has a pole of order n, at $z = z_0$, show that $b_1 = \dfrac{1}{(n-1)!}[\dfrac{d^{n-1}}{dz^{n-1}}(z-z_0)^n f(z)]_{z=z_0}$
(b) Find the pole and residue of $f(z) = \dfrac{(z+1)^2}{(z-1)^2}$

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